EMEL ÖRKEN

AI Engineer | Computer Vision | Speech AI | RAG Systems

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https://github.com/orken13 | https://orken13.github.io/emel-portfolio/

PROFESSIONAL SUMMARY

AI Engineer with hands-on production experience on H100, A100, and RTX 5090 GPUs. Delivered end-to-end projects in Computer Vision (YOLOv11/12 object detection), Speech AI (XTTS/VITS TTS synthesis, MFA acoustic modeling), and Generative AI (RAG chatbots, LangChain pipelines). Built and deployed a multi-container Docker microservice system from scratch. Strong technical foundation, fast learner, open to Junior or Part-time AI Engineer roles in software and AI companies.

EXPERIENCE

Computer Vision Engineer
Oct 2025 – Feb 2026 | Remote

TURK AI

- Trained and optimized YOLOv11n and YOLOv12n on RTX 4090 and RTX 5090 GPUs; improved mAP@50 performance through systematic hyperparameter tuning and optimizer configuration.
- Built a full end-to-end object detection pipeline: field data collection → Roboflow augmentation → model training → COCO metric evaluation (mAP@50, mAP@50–95) → inference reporting.
- Conducted manual annotation of 5,000+ real-world images in CVAT (7 classes), improving dataset quality by resolving bounding-box inconsistencies.

Stack: Python .CVAT · PyTorch · YOLOv11n/12n · OpenCV · Roboflow · Kaggle .Hugging Face

AI Engineer
Jul 2025 – Aug 2025

MANK Technologies

- Trained an XTTS model on H100 GPU with 70+ hours of audio from 30 speakers; fine-tuned VITS TTS on RTX 5090 using phonetic and non-phonetic datasets.
- Processed 9+ hours of series audio into WAV + SRT format for model training; built a text normalization pipeline to automate number-to-word conversion.
- Trained a Turkish MFA acoustic model with a custom dictionary; benchmarked Whisper Large v3, Whisper Turbo, and Phi-4 on 2.32h of data using WER/CER metrics — delivered written report to team lead.
- Developed a voice-based AI sales demo using Ultravox and prompt engineering for credit card customer service workflows.

Stack: VITS · XTTS · Whisper .Ffmpeg · MFA · Phi-4 · Ultravox · PyTorch · H100 / A100 / RTX 5090

AI Engineer Intern
Feb 2025 – Jun 2025 | Ankara

TURK AI

- Designed a 3-container Docker microservice system (RabbitMQ + PostgreSQL + MinIO) for real-time INTERPOL Red Notice data tracking; built a live Flask dashboard with filtering, sorting, and pagination.
- Transcribed YouTube videos with Whisper, embedded content into a vector database, and built a RAG chatbot supporting semantic search across multiple collections.

Stack: Docker · RabbitMQ · PostgreSQL .Websocket · MinIO · Flask · LangChain · Whisper · Vektör DB .Ffmpeg

AI Engineer Intern
Sep 2024

CBOT

- Built a RAG pipeline using FAISS and LangChain for document-based question answering.
- Developed multimodal AI applications with CLIP and BLIP; prototyped speech systems using Whisper and Tortoise TTS.
- Delivered rapid AI prototypes with Transformers, Diffusers, Gradio, and Streamlit.

Stack: LangChain · FAISS · CLIP · BLIP · Whisper · Tortoise TTS · Gradio · Streamlit .LLM .HuggingFace

NOTABLE PROJECTS

INTERPOL Real-Time Data Tracking System
3-container Docker microservice architecture · RabbitMQ message queue · PostgreSQL · MinIO object storage · Flask live dashboard with async data flow — full-stack data engineering project.

Turkish TTS Model Training — 70+ Hours of Audio
XTTS trained on H100 with 30 speakers · MFA phonetic alignment · text normalization pipeline · WER/CER benchmark report — production-grade speech synthesis system.

Production Object Detection System (TURK AI)
Real field images · YOLOv11n/12n training · Roboflow augmentation and annotation of 5,000+ images in CVAT · systematic COCO metric evaluation · RTX 5090 GPU infrastructure.

YouTube RAG Chatbot
Whisper transcription → vector embedding → multi-collection semantic search — full end-to-end RAG pipeline built from scratch.

Voice AI Sales Demo (Ultravox)
Conversational AI assistant for credit card customer service · Ultravox + prompt engineering · prototype delivered in two weeks.

TECHNICAL SKILLS

Languages:	Python .SQL .C#
Deep Learning:	PyTorch .TensorFlow · Transformers · Hugging Face
Computer Vision:	YOLOv11/v12 · OpenCV · Roboflow .CVAT
Speech AI:	VITS · XTTS · Whisper (Large v3 / Turbo) · MFA · Tortoise TTS
Generative AI:	RAG · LangChain .LangGraph · FAISS · Vector Databases · CLIP · BLIP
Infrastructure:	Docker · RabbitMQ · PostgreSQL .FastAPI.Redis.Alembic .WebSocket · MinIO · Flask
GPU Platforms:	H100 · A100 · RTX 5090 · RTX 4090
Tools:	Kaggle · Gradio · Streamlit · Diffusers

EDUCATION

B.Sc. Software Engineering *Sep 2021 – Jun 2025*
Atatürk University, Erzurum, Turkey | GPA: 3.26 / 4.00

Selçuklu Science High School *Oct 2016 – Jun 2020*
GPA: 95.2 / 100 | Ranked 2nd in school

AWARDS & ADDITIONAL

Awards: TÜBİTAK High School Research Projects — Regional Acceptance (2019) · TEV Full Scholarship (2022–2025)
Volunteer: Türk Eğitim Vakfı (TEV) Menti Program
Reference: Barış Ünver · +90 505 820 23 03 (Turk AI Founder & CEO)
Reference: Tüncer Keleş · +90 532 603 49 93 (İK Director)
Reference: Ersin Aksoy · +90 505 245 91 81 (Lotus AI Founder & CEO)
Reference: Ceyda Süer · +90 542 561 10 03 (Flexvision Tech Founder & CEO)